

Module 3 Narrative – Artifact 1 enhancement

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This artifact is the Weight Tracker App. The enhancement is to the authentication mechanism. The Weight Tracker app allows a user on android set a goal of weight loss or gain, add weights on various dates, then alert the user when the goal is reached. The app also has SMS alerts for when the goal is reached. This artifact was created in October 2025 as the project in CS-360, mobile architecture and development.

This artifact encompasses many parts of skills acquired in the computer science program. It also produces a complete, usable, practical product. It also has potential to cross into the cloud and become a full cloud application. It allows me to demonstrate my skills in application design, java programming, cloud computing and android development. The artifact was improved by changing the authentication to firebase, so it authenticates to the cloud. The app has been improved by adding an email verification, and ability to reset password via email.

I have achieved the course outcomes that I planned to meet. This enhancement is more or less complete. There may be fine-tuning as I incorporate future enhancements for other artifacts but the functionality does work as intended and firebase authentication performs properly, with edge cases being tested successfully.

Enhancing and modifying the artifact was straightforward, as I have done similar work before in architecting applications and adding in cloud authentication. However, there were things to learn in using firebase. I learned the firebase authentication process as well as session management in firebase versus a local session manager which I had previously. The main challenge was getting all of the moving pieces together in java to have firebase do the session management and update the database. I now have a better understanding of how firebase works and how it can work in my future projects.